

Module 3: Data Link Layer and Wireless Networks



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Data-Link Layer

- Data Link Control (DLC)
- Multiple Access Protocols (MAC)
- Link-Layer Addressing
- Ethernet Protocol
- Connecting Devices

Wireless Networks

- Wireless LANs
- Mobile IP

Hands-On

- Datalink Provider Interface
- SOCK_PACKET
- PF_PACKET

6.1 Wireless LANs: Introduction

Why Wireless LANs?

- Wireless communication is rapidly growing.
- Demand for cable-free connectivity is increasing.
- Wireless LANs appear in many environments.

College

Campus networks

Office

Building networks

Public Areas

Public wireless access

Wireless LAN deployment: Campus → Office → Public Area

Architectural Comparison: Medium

Wired LAN

- Uses wires to connect hosts.
- Ethernet evolved toward point-to-point access.
- Switched LAN uses a link-layer switch.
- Communication is point-to-point.
- Communication is full-duplex.

Wireless LAN

- Medium is air.
- Signal is generally broadcast.
- Hosts share the same medium.
- Medium provides multiple access.
- MAC protocols are required.

Rare Case

Point-to-point wireless is possible using limited bandwidth and two-directional antennas.

Wired point-to-point link versus wireless shared medium

Architectural Comparison: Hosts

Wired Host

- Physically connected to network.
- Fixed link-layer address.
- Address belongs to its NIC.
- Moving changes network-layer address.
- Physical connection remains necessary.

Wireless Host

- Not physically connected.
- Can move freely.
- Uses network services while moving.
- Mobility differs from wired networking.

Key Point

Wired mobility and wireless mobility are totally different issues.

Host mobility comparison: wired physical connection versus wireless free movement

Isolated LANs

Wired Isolated LAN

- Set of connected hosts.
- Uses a link-layer switch.
- Recent Ethernet uses switching.

Wireless Isolated LAN

- Called an **ad hoc network**.
- Hosts communicate freely.
- No link-layer switch exists.



Figure 6.1 *Isolated LANs: wired versus wireless*

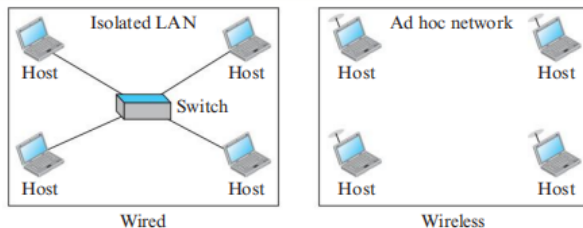
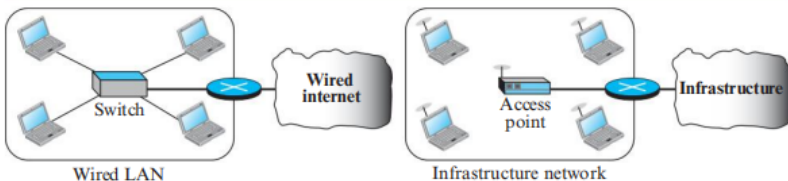


Figure 6.2 *Connection of a wired LAN and a wireless LAN to other networks*



Connection to Other Networks

Wired LAN

A wired LAN connects to another network using a **router**.

Wireless LAN

A wireless LAN may connect to:

- A wired infrastructure network.
- A wireless infrastructure network.
- Another wireless LAN.

Infrastructure Network

Wireless LAN connected to a wired infrastructure uses an **Access Point (AP)**.

Access Point and Moving Between Environments

Access Point

- Connects wireless and wired environments.
- Wireless host communicates with AP wirelessly.
- AP communicates with infrastructure using wires.
- AP role differs from a link-layer switch.

Moving Wired → Wireless

- Replace wired NIC with wireless NIC.
- Replace switch with access point.
- Link-layer address changes.
- Network-layer IP address remains unchanged.

Wired LAN → AP → Wireless LAN; IP remains, NIC/link-layer environment changes

Characteristics of Wireless LANs

Important Characteristics

- Attenuation
- Interference
- Multipath propagation
- Errors
- Access-control challenges

Why Important?

These effects are negligible or absent in many wired LAN situations.

Wireless channel problem map: Attenuation + Interference + Multipath + Error + Access

Attenuation and Interference

Attenuation

- Electromagnetic signal disperses.
- Signal strength decreases rapidly.
- Only a small portion reaches receiver.
- Battery-powered mobile senders worsen the problem.

Interference

- Receiver gets intended signal.
- Other senders may use same frequency.
- Unwanted signals interfere with reception.

Signal spreading and multiple transmitters using the same frequency band

Multipath Propagation and Error

Multipath Propagation

- Waves reflect from walls, ground, objects.
- Receiver may receive multiple signals.
- Signals travel through different paths.
- Signals arrive with different phases.
- Resulting signal becomes less recognizable.

Signal-to-Noise Ratio (SNR)

- SNR measures signal relative to noise.
- High SNR: stronger signal than noise.
- Low SNR: corrupted signal.
- Low SNR may prevent data recovery.
- Error detection, correction, retransmission become important.

Access Control: The Central Challenge

Shared Medium

Wireless hosts share the same air medium. Therefore, they need a method to access it.

Standard Ethernet

- Uses CSMA/CD.
- Host senses the medium.
- Sends when medium is idle.
- Detects collisions.
- Retransmits after collision.

Wireless Problem

Wireless LAN cannot reliably use collision detection.

CSMA/CD: Sense → Transmit → Detect Collision

Why CSMA/CD Fails in Wireless LANs

❶ Cannot transmit and receive simultaneously

- Collision detection requires duplex operation.
- Battery-powered hosts cannot support this efficiently.

❷ Hidden station problem

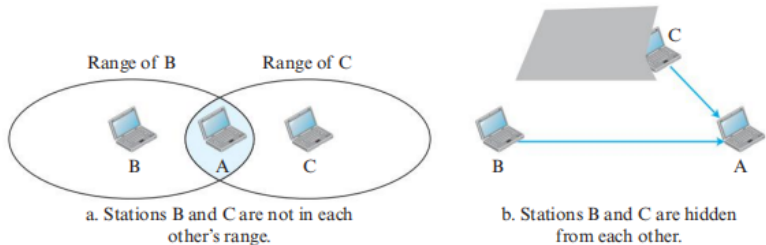
- Stations may not hear each other.
- Collision can occur without detection.

❸ Distance and signal fading

- Stations may be far apart.
- Collision signal may not reach the sender.

Three reasons CSMA/CD is unsuitable for wireless LANs

Figure 6.3 *Hidden station problem*



Hidden Station Problem

Scenario

- Station B can reach station A.
- Station C can reach station A.
- B and C cannot hear each other.
- B sends data to A.
- C senses the medium as idle.
- C sends data to A.
- Collision occurs at A.

Definition

B and C are **hidden from each other** with respect to A.

CSMA/CA: Wireless Solution

Carrier Sense Multiple Access with Collision Avoidance

- Designed for wireless LANs.
- Collisions cannot be reliably detected.
- Therefore, collisions must be avoided.

IFS

Wait before transmitting

Contention Window

Random waiting slots

ACK

Confirm reception

6.1.2 IEEE 802.11 Project

IEEE 802.11

- Specifies wireless LAN physical layer.
- Specifies wireless LAN data-link layer.

WiFi

- Common term for wireless LAN.
- WiFi means “wireless fidelity”.
- Certified by the WiFi Alliance.
- Global nonprofit industry association.
- More than 300 member companies.
- Promotes wireless LAN growth.

IEEE 802.11 protocol stack and WiFi certification relationship

Basic Service Set (BSS)

Definition

BSS is a basic building block of an IEEE 802.11 wireless LAN.

Components

- Stationary wireless stations.
- Mobile wireless stations.
- Optional Access Point (AP).

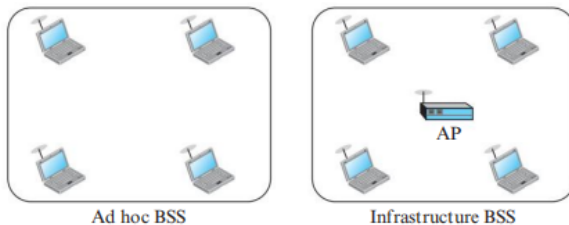
Without AP

- Stand-alone network.
- Ad hoc architecture.
- Cannot communicate with other BSSs.

With AP

Infrastructure BSS.

Figure 6.4 *Basic service sets (BSSs)*



Extended Service Set (ESS)

Definition

ESS contains two or more BSSs with APs.

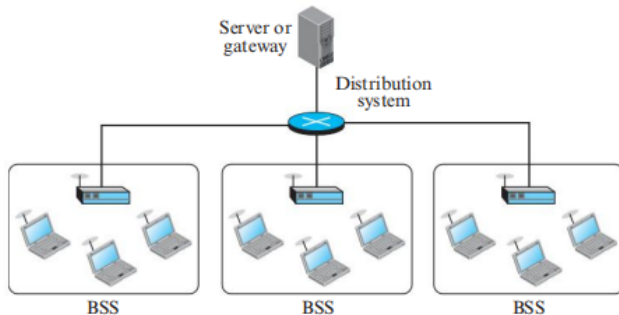
Distribution System

- Connects APs in different BSSs.
- May be wired or wireless.
- IEEE 802.11 does not restrict its technology.
- Can use an IEEE LAN such as Ethernet.

Station Types

- Mobile stations inside BSSs.
- Stationary AP stations in wired LAN.

Figure 6.5 *Extended service set (ESS)*



BSS Communication and Mobility

Communication

- Stations within range can communicate directly.
- Outside-BSS communication uses the AP.
- BSS resembles a cellular network cell.
- AP resembles a cellular base station.

Important

A mobile station can belong to more than one BSS simultaneously.

ESS analogy: BSS = Cell; AP = Base Station

Station Mobility Types

No-Transition

- Stationary, or
- Moves inside one BSS.

BSS-Transition

- Moves between BSSs.
- Remains inside one ESS.

ESS-Transition

Moves from one ESS to another ESS.

Limitation

IEEE 802.11 does not guarantee continuous communication during movement.

Mobility path: Inside BSS → Across BSSs → Across ESSs

IEEE 802.11 MAC Sublayer

Two MAC Sublayers

DCF

Distributed Coordination Function

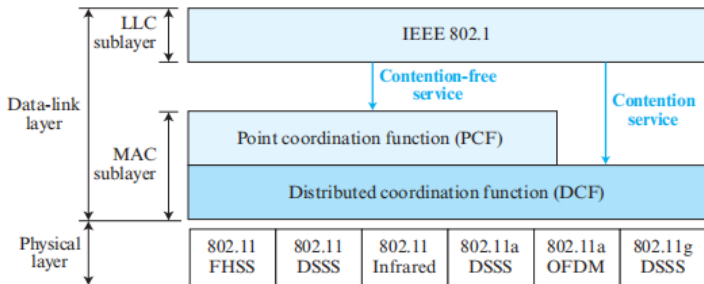
- Distributed access.
- Uses CSMA/CA.

PCF

Point Coordination Function

- Optional.
- Centralized polling.

Figure 6.6 *MAC layers in IEEE 802.11 standard*



DCF and CSMA/CA

Distributed Coordination Function

- One MAC protocol defined by IEEE 802.11.
- Uses CSMA/CA as access method.

Why CSMA/CA?

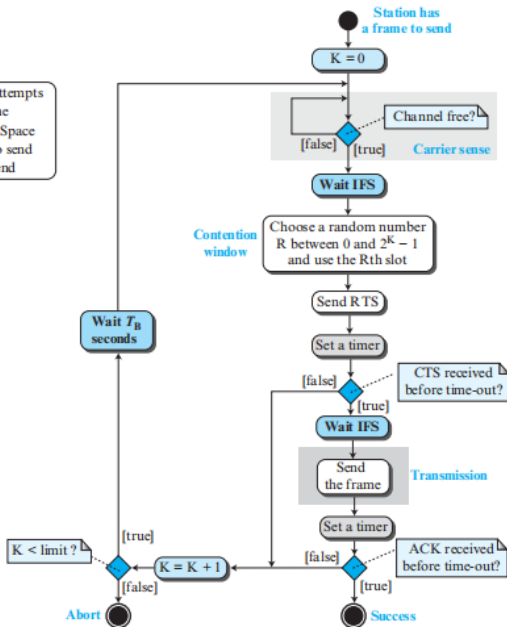
- Wireless collisions cannot be reliably detected.
- Avoidance is therefore necessary.
- Uses IFS, contention window and acknowledgments.

DCF $\rightarrow$ CSMA/CA $\rightarrow$ IFS + Contention Window + ACK

Figure 6.7 Flow diagram of CSMA/CA

Legend

K: Number of attempts
 T_B : Back-off time
 IFS: Interframe Space
 RTS: Request to send
 CTS: Clear to send
 CTS: Clear to send



CSMA/CA: Three Strategies

1. Interframe Space (IFS)

- Station waits even when channel appears idle.
- Distant transmission may not have arrived yet.
- IFS allows signal front to arrive.
- Shorter IFS can provide higher priority.
- Contention time still follows IFS.

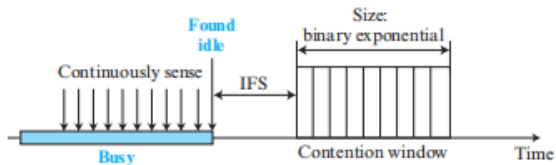
2. Contention Window

- Divided into time slots.
- Station chooses random waiting slots.
- Binary exponential backoff changes window size.
- Starts at one slot.
- Doubles after failure.
- Channel is sensed after every slot.
- Busy channel pauses timer; timer resumes later.
- Longest waiting station receives priority.

3. Acknowledgment

- Collision or corruption may still occur.
- Positive ACK confirms successful reception.
- Timeout indicates missing ACK.

Figure 6.8 Contention window



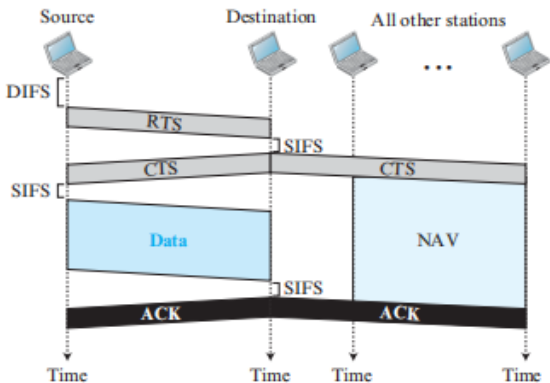
Frame Exchange Timeline

- 1 Source senses carrier energy.
- 2 Persistence/backoff continues until channel is idle.
- 3 Waits DIFS.
- 4 Sends RTS.
- 5 Destination waits SIFS.
- 6 Destination sends CTS.
- 7 Source waits SIFS.
- 8 Source sends DATA.
- 9 Destination waits SIFS.
- 10 Destination sends ACK.

Why ACK?

Wireless station cannot directly determine successful data arrival as CSMA/CD can.

Figure 6.9 CSMA/CA and NAV



NAV and Collision During Handshaking

Network Allocation Vector (NAV)

- RTS includes required channel duration.
- Other stations create NAV timers.
- NAV shows remaining busy period.
- Station checks NAV before physical sensing.
- Expired NAV permits channel checking.

Collision During RTS/CTS

- Multiple stations may send RTS simultaneously.
- RTS frames may collide.
- Collision cannot be directly detected.
- No CTS implies collision.
- Sender applies backoff and retries.

Hidden Station Solution: RTS/CTS

Problem

B's transmission reaches A but not C. C cannot hear B.

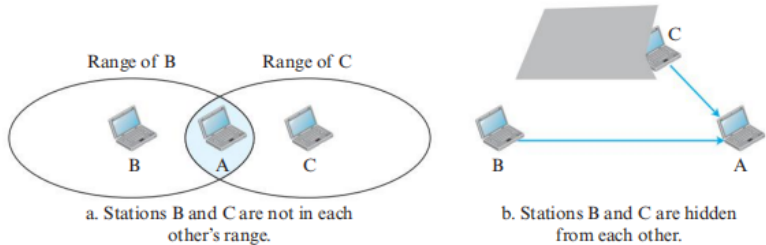
Solution

- 1 B sends RTS to A.
- 2 A sends CTS to B.
- 3 CTS contains transmission duration.
- 4 CTS reaches C.
- 5 C learns the channel is reserved.
- 6 C defers transmission.

Result

RTS/CTS reduces hidden-station collisions.

Figure 6.3 *Hidden station problem*



Point Coordination Function (PCF)

PCF Characteristics

- Optional access method.
- Only for infrastructure networks.
- Not used in ad hoc networks.
- Implemented on top of DCF.
- Mainly for time-sensitive transmission.

Centralized Polling

- AP performs polling.
- Polls stations one after another.
- Polled stations send data to AP.

PCF centralized polling sequence: AP/Point Controller → Station 1 → Station 2 → Station 3

PCF Priority, Repetition Interval and Fragmentation

PIFS

- PIFS = PCF Interframe Space.
- PIFS is shorter than DIFS.
- Therefore PCF gets priority over DCF.

Repetition Interval

- Contains contention-free PCF period.
- Also contains contention-based DCF traffic.
- Starts with a beacon frame.
- Stations set NAV for contention-free period.
- PC can poll, receive data, send/receive ACK.
- Piggybacking is supported.
- CF End releases the medium.

Fragmentation

Noisy wireless links corrupt frames. Large frames are divided into smaller frames. Retransmitting smaller frames is more efficient.

MAC Frame Format and Frame Types

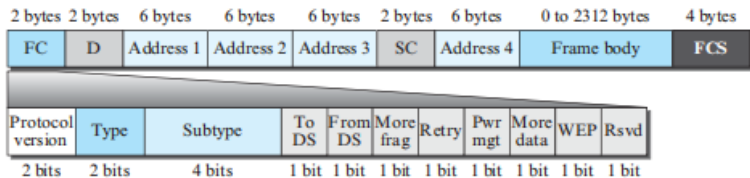
IEEE 802.11 MAC Frame: Nine Fields

- **Frame Control (FC):** 2 bytes.
- **D:** duration; sets NAV or frame ID.
- **Addresses:** four fields, 6 bytes each.
- **Sequence Control (SC):** 16 bits.
- First 4 bits = fragment number.
- Last 12 bits = sequence number.
- **Frame Body:** 0–2312 bytes.
- **FCS:** 4 bytes, CRC-32.

Three Frame Categories

- **Management:** initial station/AP communication.
- **Control:** channel access and acknowledgments.
- **Data:** data and control information.

Figure 6.11 *Frame format*



Addressing Mechanism and Exposed Station

Four Addressing Cases

To DS	From DS	Address 1	Address 2	Address 3
0	0	Destination	Source	BSS ID
0	1	Destination	Sending AP	Source
1	0	Receiving AP	Source	Destination
1	1	Receiving AP	Sending AP	Destination

Address Rules

- Address 1 = next device.
- Address 2 = previous device.
- Address 3 = final destination or original source.
- Address 4 = original source when DS is wireless.

Exposed Station

C hears A transmitting to B. C therefore defers, although C could transmit to D. RTS/CTS does not solve this problem.

Figure 6.12 *Control frames*



Figure 6.13 Addressing mechanisms

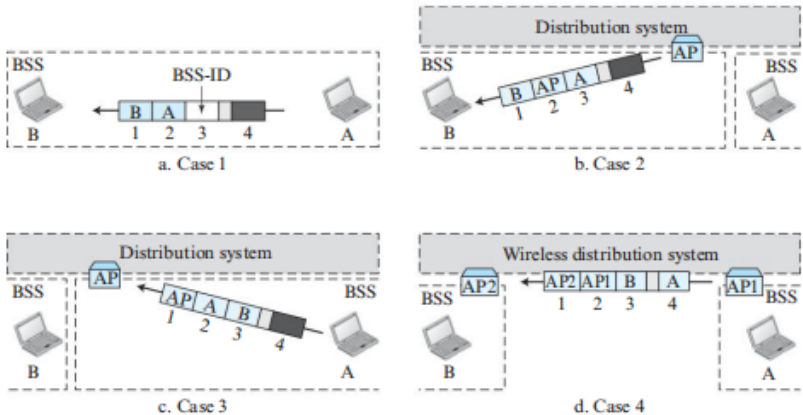


Figure 6.14 *Exposed station problem*

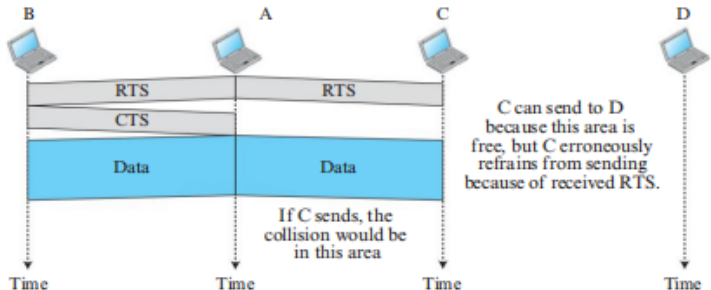
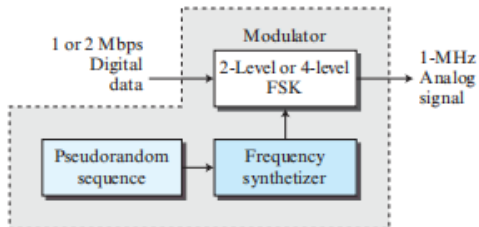


Figure 6.15 *Physical layer of IEEE 802.11 FHSS*



Physical Layer Specifications and 802.11 Versions

ISM Bands

All except infrared use:

- 902–928 MHz
- 2.400–4.835 GHz
- 5.725–5.850 GHz

IEEE	Technique	Band	Rate
802.11	FHSS	2.400–4.835 GHz	1, 2 Mbps
802.11	DSSS	2.400–4.835 GHz	1, 2 Mbps
None	Infrared	800–950 nm	1, 2 Mbps
802.11a	OFDM	5.725–5.850 GHz	6–54 Mbps
802.11b	DSSS	2.400–4.835 GHz	5.5, 11 Mbps
802.11g	OFDM	2.400–4.835 GHz	22, 54 Mbps
802.11n	OFDM	5.725–5.850 GHz	600 Mbps

Figure 6.16 *Physical layer of IEEE 802.11 DSSS*

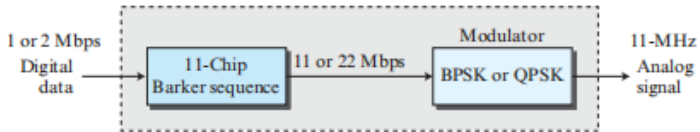


Figure 6.17 *Physical layer of IEEE 802.11 infrared*

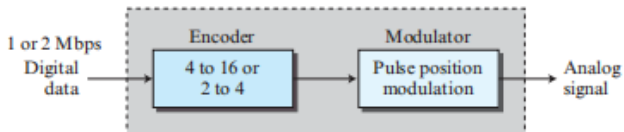
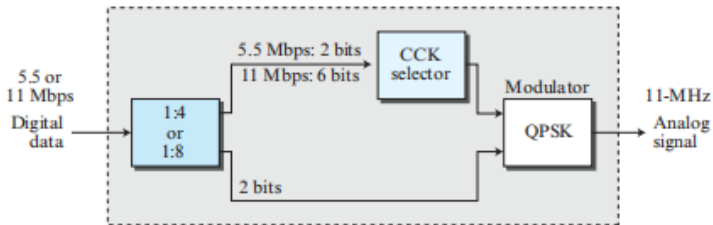


Figure 6.18 *Physical layer of IEEE 802.11b*



Bluetooth: Short-Range Wireless Networking

What is Bluetooth?

- Wireless LAN technology designed to connect devices at a **short distance**.
- Devices may include:
 - Telephones
 - Notebooks and computers
 - Cameras
 - Printers
 - Other electronic gadgets

Network Characteristics

- Bluetooth LAN is an **ad hoc network**.
- Devices spontaneously find one another.
- The resulting small network is called a **piconet**.
- A Bluetooth LAN can connect to the Internet if one gadget provides that capability.
- It is inherently limited in size.

Bluetooth: Applications and Origin

Applications

- Wireless mouse and keyboard communication.
- Monitoring devices communicating with sensors.
- Home security sensors connected to a security controller.
- Synchronization of laptop computers at conferences.

Origin

- Bluetooth was originally started by **Ericsson**.
- Named after **Harald Blaatand**, king of Denmark.
- Blaatand united Denmark and Norway.
- “Blaatand” translates to **Bluetooth**.

Standard

- Bluetooth technology implements a protocol defined by **IEEE 802.15**.
- Designed for a wireless **personal-area network (PAN)**.

Bluetooth Architecture: Piconet and Scatternet

Two Bluetooth Network Types

- **Piconet:** a small Bluetooth network.
- **Scatternet:** combination of multiple piconets.

Piconet

- Maximum of **eight active stations**.
- One station is the **primary**.
- Remaining stations are **secondaries**.
- Secondaries synchronize their clocks and hopping sequence with the primary.

Scatternet

- Formed by combining piconets.
- A secondary in one piconet can act as primary in another.
- A station can belong to **two piconets**.

Figure 6.19 *Piconet*

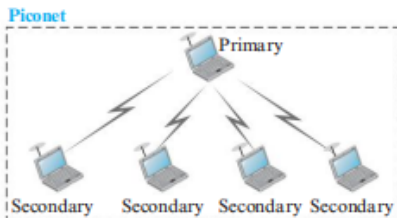
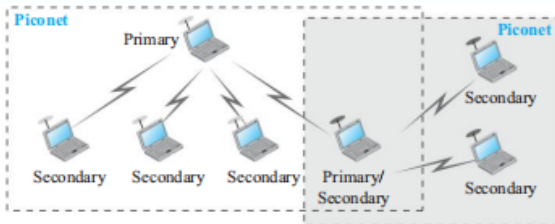


Figure 6.20 *Scatternet*



Piconet: Primary, Secondaries and Parked State

Active Stations

- One primary station.
- Up to seven active secondary stations.
- Communication can be:
 - One-to-one
 - One-to-many

Parked State

- Additional secondaries can remain in the **parked state**.
- Parked stations remain synchronized with the primary.
- They cannot participate in communication.
- To activate a parked station, an active station must move to the parked state.
- Only eight stations can be active simultaneously.

Scatternet: Connecting Piconets

Basic Idea

A **scatternet** is created when piconets are combined.

- A secondary station in one piconet can become the primary of another piconet.
- The same station can receive messages from the primary of the first piconet.
- It can then act as a primary and deliver messages to secondaries in another piconet.
- A station can therefore be a member of **two piconets**.

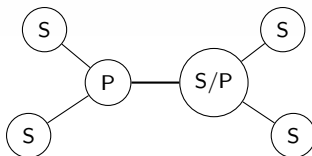
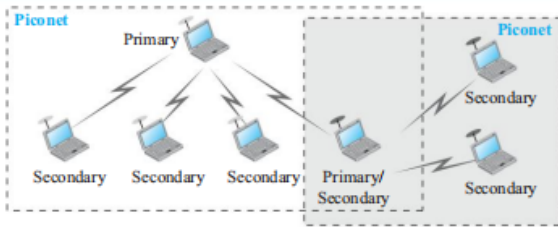


Figure 6.20 *Scatternet*



Bluetooth Devices: Rate, Band and Interference

Bluetooth Device

- Contains a built-in **short-range radio transmitter**.
- Current data rate given in the text: **1 Mbps**.
- Uses a **2.4-GHz** band.
- Intended for short-range communication.

Interference

- Bluetooth and IEEE 802.11b WLANs can operate in overlapping frequency ranges.
- Therefore, interference between Bluetooth LANs and 802.11b WLANs is possible.

Think

Why can two independent wireless networks interfere even though they belong to different technologies?

Bluetooth Protocol Layers

Protocol Architecture

Bluetooth uses several layers that do not exactly match the Internet model.

- **L2CAP** – roughly equivalent to the LLC sublayer in LANs.
- **Baseband** – roughly equivalent to the MAC sublayer.
- **Radio** – roughly equivalent to the physical layer.

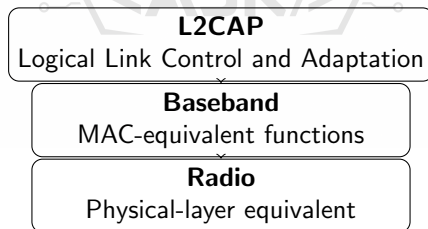
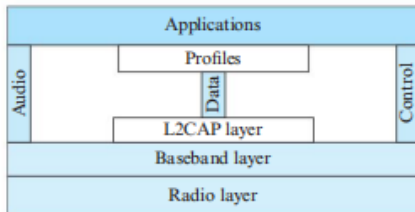


Figure 6.21 *Bluetooth layers*



L2CAP: Logical Link Control and Adaptation Protocol

Purpose

L2CAP is roughly equivalent to the **LLC sublayer** in LANs.

- Used for data exchange on an **ACL link**.
- **SCO channels do not use L2CAP**.
- Creates a virtual channel for upper-layer protocols.

Field	Description
Length	16-bit field defining the size of data from upper layers
Maximum data	Up to 65,535 bytes
CID	Channel Identifier; identifies the virtual channel

Figure 6.22 *L2CAP data packet format*



L2CAP Functions: Multiplexing and Segmentation

Multiplexing

- Sender accepts data from an upper-layer protocol.
- Frames the data.
- Delivers it to the baseband layer.
- Receiver accepts a frame from baseband.
- Extracts the data and delivers it to the appropriate protocol layer.
- Creates a type of virtual channel.

Segmentation and Reassembly

- Baseband payload maximum: **2774 bits = 343 bytes.**
- 4 bytes are needed for packet definition and length.
- Maximum upper-layer packet that fits directly: **339 bytes.**
- L2CAP can handle packets up to **65,535 bytes.**
- Large packets are segmented at source and reassembled at destination.

L2CAP Functions: QoS and Group Management

Quality of Service (QoS)

- Bluetooth allows stations to define a QoS level.
- If no QoS level is defined, Bluetooth uses **best-effort service**.
- Best effort means the system does its best under the circumstances.

Group Management

- Allows devices to create a type of logical addressing.
- Similar to **multicasting**.
- Two or three secondary devices can form a multicast group.
- The primary can send data to the group.

Remember

L2CAP = **Multiplexing + Segmentation/Reassembly + QoS + Group Management**

Baseband Layer: TDD-TDMA

Baseband Layer

- Roughly equivalent to the **MAC sublayer**.
- Bluetooth uses **TDMA**.
- Primary and secondary stations communicate using time slots.
- Slot length = dwell time = **625 μ s**.

TDD-TDMA

- TDD = Time-Division Duplex.
- Communication is **half-duplex**.
- Sender and receiver do not transmit simultaneously.
- The two directions use different hops.
- Similar to walkie-talkies using different carrier frequencies.

Bluetooth TDMA: Single and Multiple Secondaries

Single Secondary

- Slots are $625\ \mu\text{s}$.
- Primary uses even slots:

0, 2, 4, ...

- Secondary uses odd slots:

1, 3, 5, ...

- Slot 0: primary $\rightarrow$ secondary.
- Slot 1: secondary $\rightarrow$ primary.

Multiple Secondaries

- Primary still uses even-numbered slots.
- All secondaries listen during even slots.
- A secondary transmits in the next odd slot if the previous frame was addressed to it.
- Other secondaries remain silent.
- This resembles **poll/select with reservations**.

Figure 6.23 *Single-secondary communication*

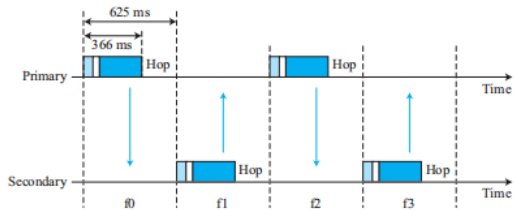
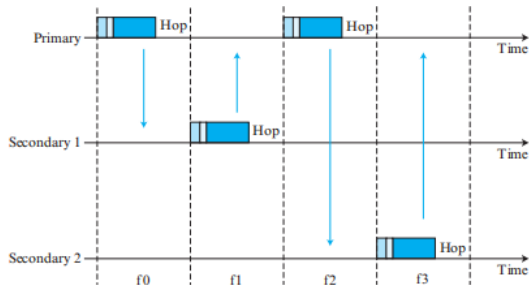


Figure 6.24 *Multiple-secondary communication*



Multiple-Secondary Communication: Example

- ➊ In **slot 0**, the primary sends a frame to secondary 1.
- ➋ In **slot 1**, only secondary 1 sends a frame to the primary.
- ➌ In **slot 2**, the primary sends a frame to secondary 2.
- ➍ In **slot 3**, only secondary 2 sends a frame to the primary.
- ➎ The cycle continues.

Key Idea

When the primary selects a secondary, it also effectively **polls** it. The next time slot is reserved for the selected secondary to send.

- If the selected secondary has data → it transmits.
- If it has no data → the channel remains silent.

Bluetooth Links: SCO vs ACL

Two Types of Bluetooth Links

Bluetooth supports two types of links between a primary and a secondary: **SCO** and **ACL**.

Feature	SCO	ACL
Full name	Synchronous Connection-Oriented	Asynchronous Connectionless
Priority	Avoiding latency is more important	Data integrity is more important
Retransmission	Damaged packets are not retransmitted	Corrupted payloads are retransmitted
Slot usage	Specific slots are reserved at regular intervals	Uses available odd-numbered slots
Basic operation	Physical link with reserved slots	Connectionless data transfer
Main use	Real-time audio	Data communication
Data rate	64 kbps per link	Maximum 721 kbps
Links	Up to three SCO links with the primary	Can use one, three, or more slots

Bluetooth Baseband Frame Formats

Three Frame Types

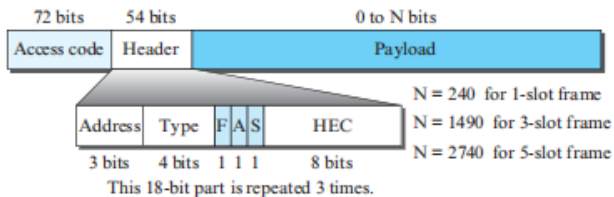
A baseband frame can be:

- **One-slot frame**
- **Three-slot frame**
- **Five-slot frame**

Frame	Slot duration	Effective duration	Size
One-slot	$625 \mu s$	$625 - 259 = 366 \mu s$	366 bits
Three-slot	$3 \times 625 \mu s$	$1875 - 259 = 1616 \mu s$	1616 bits
Five-slot	$5 \times 625 \mu s$	$3125 - 259 = 2866 \mu s$	2866 bits

- $259 \mu s$ is used for hopping and control mechanisms.
- Three-slot and five-slot frames remain at the same hop for multiple slots.
- Three hop numbers are consumed by a three-slot frame; its hop number corresponds to the first slot.

Figure 6.25 *Frame format types*



Bluetooth Baseband Frame Fields

Field	Description
Access Code	72-bit field containing synchronization information and the primary identifier.
Header	54-bit field formed by three repeated 18-bit patterns.
Address	3 bits; identifies up to seven secondaries. Address 0 supports broadcast from primary.
Type	4 bits; identifies the type of data from upper layers.
F	1-bit flow-control field. 1 indicates that the device cannot receive more frames.
A	1-bit acknowledgment field. Bluetooth uses Stop-and-Wait ARQ.
S	1-bit sequence-number field.
HEC	8-bit header error-correction/checksum field.
Payload	0–2740 bits; carries data or control information.

Header Error Control

The three identical 18-bit header sections are compared bit by bit. If corresponding bits disagree, the majority value is used. This provides forward error correction for the header.

Radio Layer: FHSS and GFSK

Radio Layer

- Roughly equivalent to the physical layer.
- Bluetooth devices are low-power.
- Range is approximately **10 m**.
- Uses the **2.4-GHz ISM band**.
- Band divided into **79 channels**, each 1 MHz.

FHSS

- Uses **Frequency-Hopping Spread Spectrum**.
- Helps avoid interference from other devices/networks.
- Hops **1600 times per second**.
- Each frequency is used for:

$$\frac{1}{1600} s = 625 \mu s$$

- 625 μs is the **dwelt time**.

GFSK Modulation

Bluetooth uses **GFSK**: FSK with Gaussian bandwidth filtering. Bit 1 is represented by frequency deviation above the carrier; bit 0 by frequency deviation

WiMAX: Worldwide Interoperability for Microwave Access

Purpose

WiMAX is an IEEE wireless standard designed to provide a “last mile” broadband wireless alternative to:

- Cable modem
- Telephone DSL service
- Existing telecommunications infrastructures

Standards

- IEEE **802.16**: fixed wireless.
- IEEE **802.16e**: mobile wireless.
- 802.16e-2005 specifies 2–11 GHz operation, SOFDMA, MIMO and full mobility support.

Coverage

- Supports line-of-sight (LOS) subscribers.
- Also supports near/non-line-of-sight (NLOS) subscribers.
- Text compares approximately:

Radio and Antenna

- Basic units are the **radio** and **antenna**.
- Each radio contains transmitter and receiver.
- Signals operate between **2 and 11 GHz**.
- WiMAX uses **Software Defined Radios (SDR)**.

Antenna Types

- Omni-directional
- Sector
- Panel

WiMAX also uses a **beamsteering adaptive antenna system (AAS)**.

- Transmission energy can be focused toward the receiver.
- Reception can be focused toward the transmitter.

WiMAX: Interference Avoidance and Throughput

Key Technologies

WiMAX uses several technologies to improve performance:

- **OFDMA** – multiple-access method allowing simultaneous transmissions to and from several users.
- **MIMO** – multiple-input/multiple-output antenna system.
- **AAS** – adaptive antenna system for beam steering.

OFDMA

- Supports multiple simultaneous users.
- Works well with AAS and MIMO.
- Helps improve throughput.

Combined Effect

The text states that these techniques can:

- Significantly improve throughput.
- Increase link range.
- Reduce interference.

WiMAX Subscriber Stations

Customer Premises Equipment (CPE)

The subscriber station is also called the **subscriber unit**.

Indoor Unit

- Approximately the size of a cable/DSL modem.
- Can be self-installed.
- Radio losses require the subscriber to be closer to the base station.

Outdoor Unit

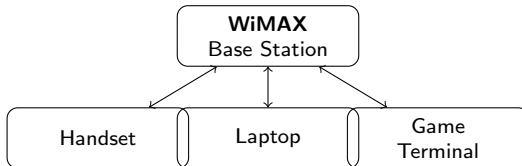
- Approximately the size of a residential satellite dish.
- Requires professional installation.

WiMAX Portable Units

Mobile WiMAX and Portable Devices

With the potential for mobile WiMAX, increasing attention is given to portable units.

- Handsets
- PC peripherals
- Embedded devices in laptops
- Consumer electronic devices
- Game terminals
- MP3 players and similar devices



WiMAX Data-Link Layer: Scheduling vs Contention

WiFi

- MAC uses **contention access**.
- Subscriber stations compete for access.
- Stations farther from the AP can be repeatedly interrupted by closer stations.

WiMAX

- MAC uses a **scheduling algorithm**.
- Subscriber station competes only once for initial network entry.
- An access slot is then allocated to the subscriber.

Key Difference

WiFi: Contention-based access

WiMAX: Scheduled access

WiMAX Physical Layer

IEEE 802.16e-2005

The physical layer specification includes:

- **2–11 GHz** frequency range.
- **Scalable OFDMA (SOFDMA)**.
- **MIMO antenna** technology.
- Capability for **full mobility support**.

SOFDMA

- Scalable OFDMA.
- Provides multiple-access capability.
- Supports simultaneous users.

MIMO

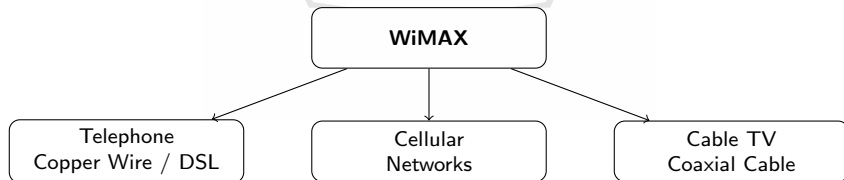
- Multiple-input/multiple-output antenna system.
- Used as part of WiMAX's physical-layer technology.
- Supports the goal of improved wireless performance.

WiMAX: Applications and Infrastructure Alternatives

Main Goal

WiMAX aims to provide cost-effective alternatives to several existing telecommunications infrastructures.

- Telephone company's **copper wire**.
- Cellular network infrastructure.
- Cable TV's **coaxial cable** infrastructure.



Bluetooth vs WiMAX: Conceptual Comparison

Feature	Bluetooth	WiMAX
Network focus	Short-range wireless networking	Broadband wireless access
Architecture	Ad hoc piconets / scatter-nets	Base-station infrastructure
Typical scope	Personal-area / short range	Large-area / last-mile access
Access	TDD-TDMA	Scheduling algorithm
Key technologies	FHSS, GFSK, TDMA	OFDMA, SOFDMA, MIMO, AAS
Typical role	Connecting nearby gadgets	Broadband connectivity

Important

The two technologies address different wireless networking requirements: Bluetooth emphasizes short-range device connectivity, while WiMAX targets broadband wireless access.

Key Takeaways

Bluetooth

- Short-range wireless networking.
- Ad hoc architecture.
- Piconet has one primary and up to seven active secondaries.
- Multiple piconets can form a scatternet.
- L2CAP provides multiplexing, segmentation/reassembly, QoS and group management.
- Baseband uses TDD-TDMA.
- SCO prioritizes low latency; ACL prioritizes data integrity.
- Radio uses 2.4-GHz ISM, FHSS

WiMAX

- Designed for broadband wireless “last mile” access.
- Uses base-station infrastructure.
- Includes subscriber and portable units.
- MAC uses scheduling rather than contention access.
- Uses AAS, OFDMA and MIMO.
- 802.16e-2005 specifies 2–11 GHz, SOFDMA, MIMO and full mobility support.

Classroom Case Study / Discussion

Scenario

A college wants wireless connectivity for three situations:

- 1 Students connecting wireless keyboards, mice and nearby gadgets.
- 2 Sensors communicating with a small campus security controller.
- 3 Providing broadband wireless connectivity across a large campus area.

Discuss

- Which Bluetooth architecture would be involved in the first two situations?
- Why is a piconet limited to eight active stations?
- Why does Bluetooth use TDD-TDMA?
- When would SCO be preferred over ACL?
- How does WiMAX scheduling differ from WiFi contention access?
- How do AAS, OFDMA/SOFDMA and MIMO contribute to WiMAX operation?